

Lab4: Insight into PHP and SQL

Objective

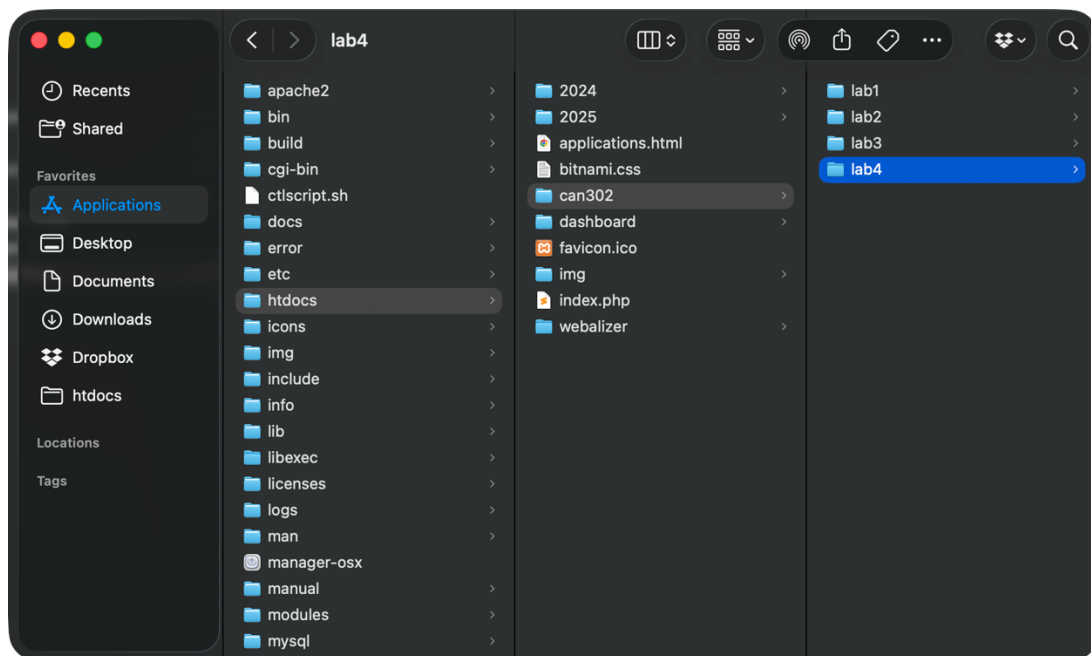
- Demonstrate the basic working knowledge of PHP
- Utilize SQL to create, query, and manage databases
- Integrate SQL with PHP to build dynamic web pages

Tips

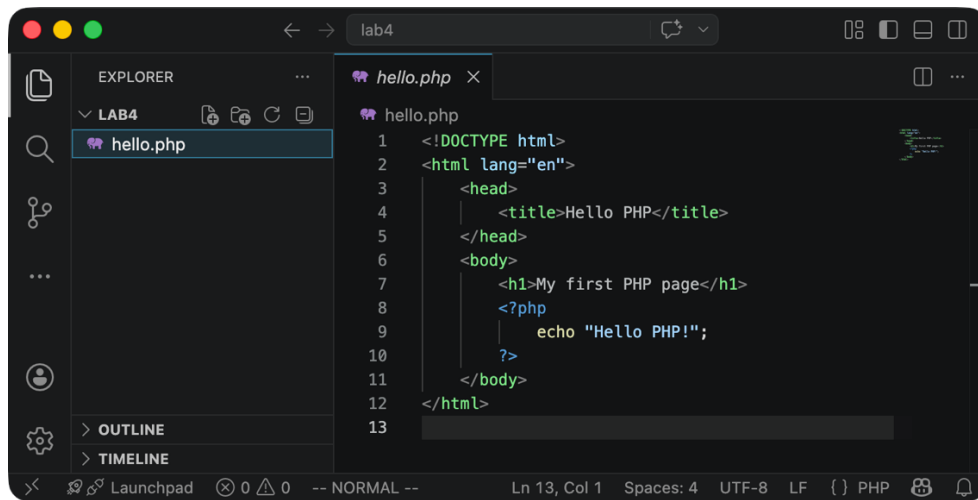
- These labs are expected to take more than the allocated 2 hours. You should try the tasks before the lab. Practice makes code perfect.
- Before the lab, you could post your questions in LMO forum, and you are welcome to offer solutions to the questions in LMO.
- During the lab, you could discuss with TAs for your questions. Face to face discussion will have a relatively high efficiency.

A. PHP Basics

1. Create a new subfolder "lab4" under folder "can 302" (The following screenshot is in macOS, but same operations can be applied to Windows):

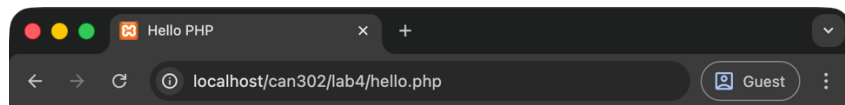


2. Use VS Code to open "lab4" and create a new file "hello.php" as follows:



```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Hello PHP</title>
5   </head>
6   <body>
7     <h1>My first PHP page</h1>
8     <?php
9       echo "Hello PHP!";
10    ?>
11   </body>
12 </html>
13
```

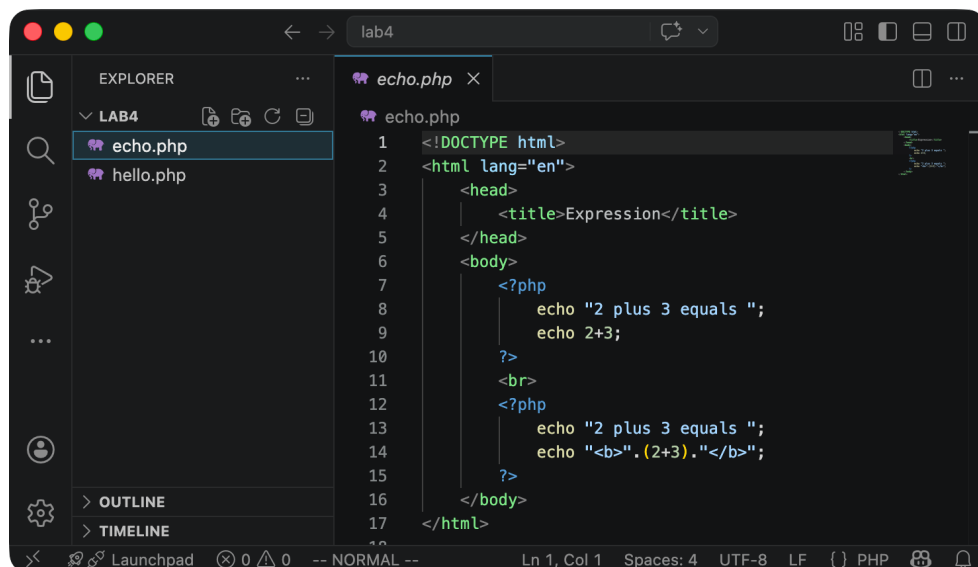
3. Access the webpage in your browser via URL: <http://localhost/can302/lab4/hello.php>



My first PHP page

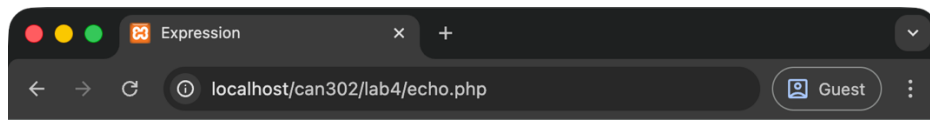
Hello PHP!

4. Create a new file "echo.php" under "lab4" as follows:

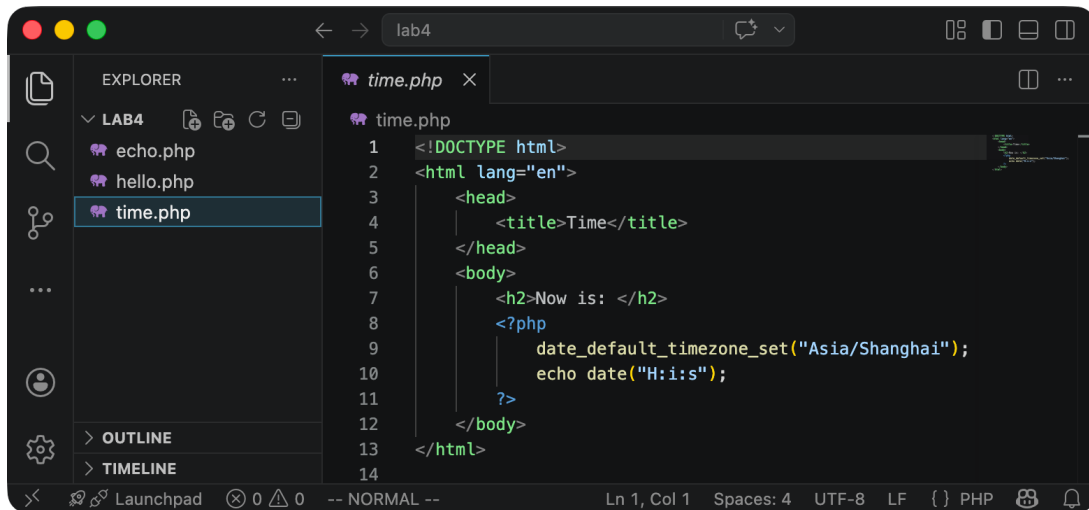


```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Expression</title>
5   </head>
6   <body>
7     <?php
8       echo "2 plus 3 equals ";
9       echo 2+3;
10    ?>
11     <br>
12     <?php
13       echo "2 plus 3 equals ";
14       echo "<b>".(2+3)."</b>";
15     ?>
16   </body>
17 </html>
18
```

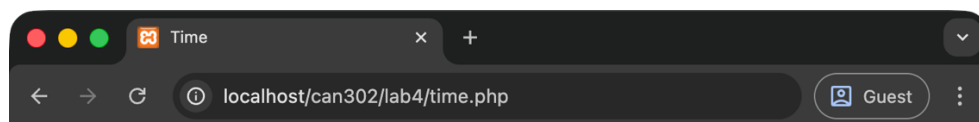
5. Access the webpage in your browser via URL: <http://localhost/can302/lab4/echo.php>



6. Create a new file "time.php" under "lab4" as follows:



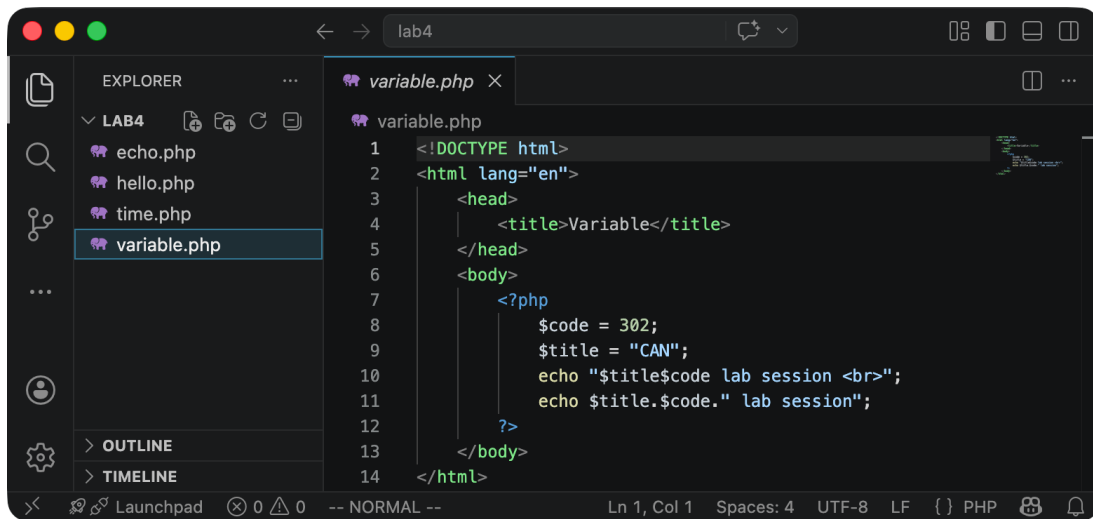
7. Access the webpage in your browser via URL: <http://localhost/can302/lab4/time.php>



Now is:

15:17:12

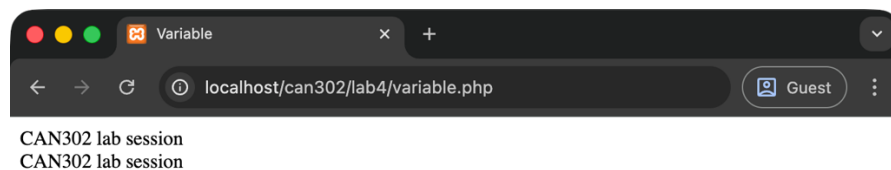
8. Create a new file "variable.php" under "lab4" as follows:



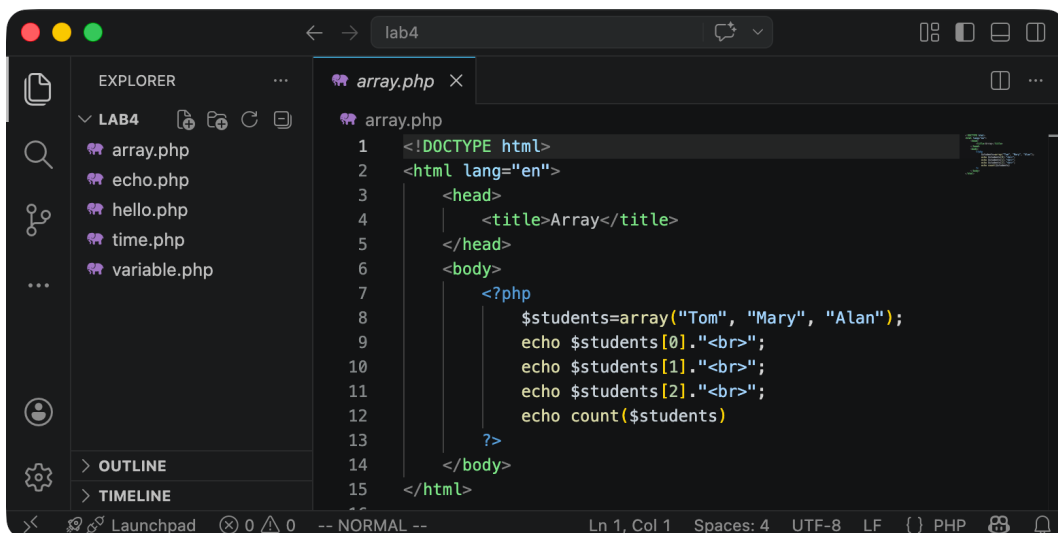
The screenshot shows a code editor with a file explorer on the left. The file explorer shows a directory named 'LAB4' containing files: 'echo.php', 'hello.php', 'time.php', and 'variable.php'. The 'variable.php' file is selected. The main editor area shows the following code:

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Variable</title>
5   </head>
6   <body>
7     <?php
8       $code = 302;
9       $title = "CAN";
10      echo "$title$code lab session <br>";
11      echo $title.$code." lab session";
12    ?>
13  </body>
14 </html>
```

9. Access the webpage in your browser via URL: <http://localhost/can302/lab4/variable.php>



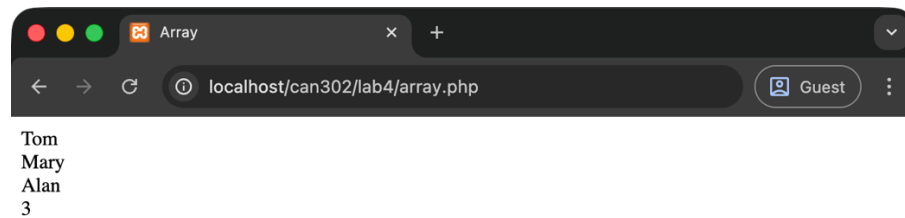
10. Create a new file "array.php" under "lab4" as follows:



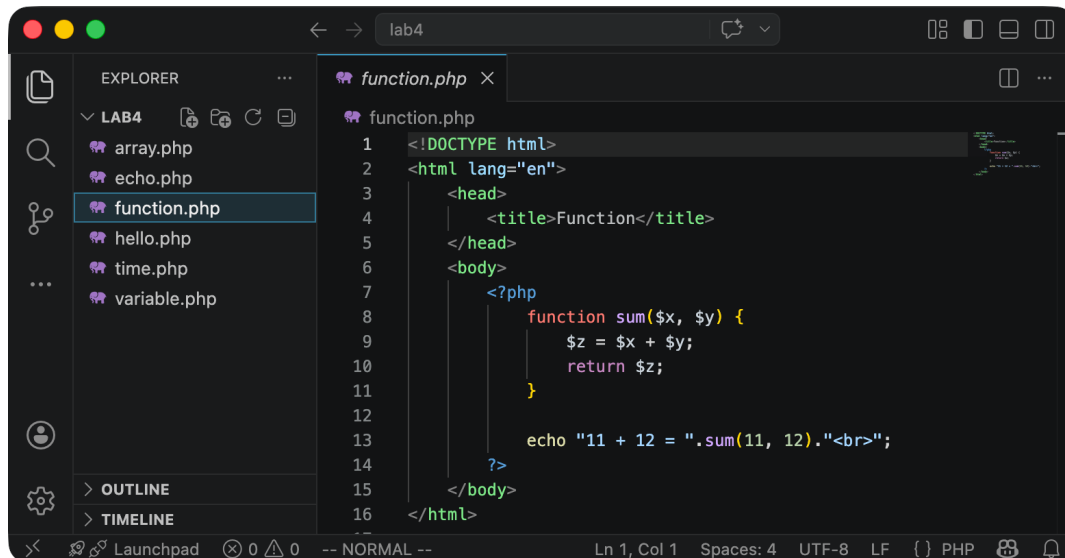
The screenshot shows a code editor with a file explorer on the left. The file explorer shows a directory named 'LAB4' containing files: 'array.php', 'echo.php', 'hello.php', 'time.php', and 'variable.php'. The 'array.php' file is selected. The main editor area shows the following code:

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Array</title>
5   </head>
6   <body>
7     <?php
8       $students=array("Tom", "Mary", "Alan");
9       echo $students[0]."<br>";
10      echo $students[1]."<br>";
11      echo $students[2]."<br>";
12      echo count($students)
13    ?>
14  </body>
15 </html>
```

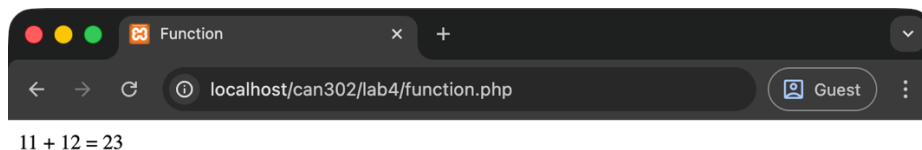
11. Access the webpage in your browser via URL: <http://localhost/can302/lab4/array.php>



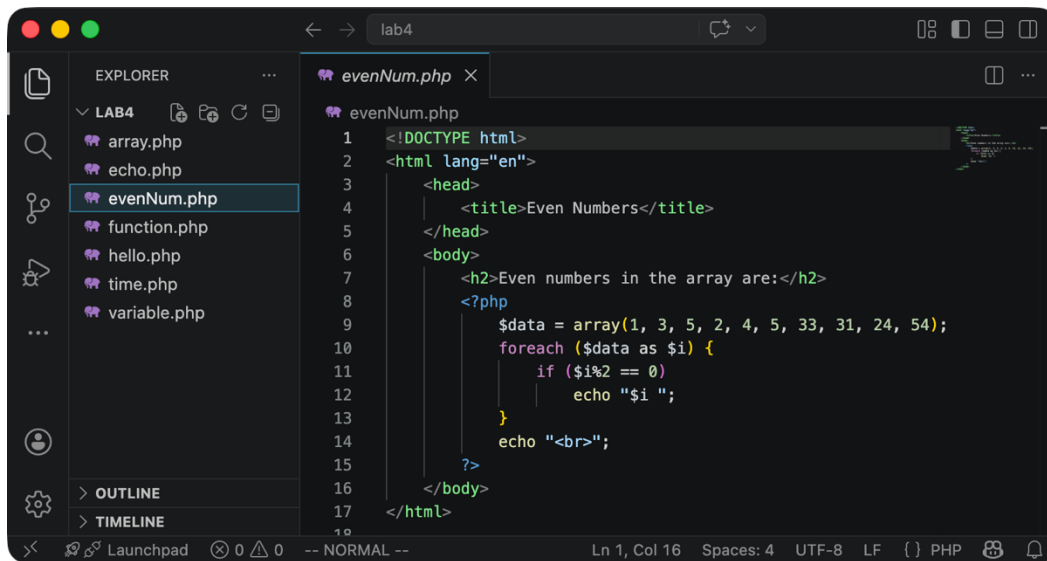
12. Create a new file "function.php" under "lab4" as follows:



13. Access the webpage in your browser via URL: <http://localhost/can302/lab4/function.php>

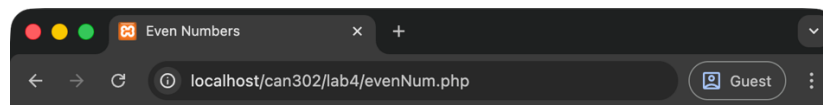


14. Create a new file "evenNum.php" under "lab4" as follows:



```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Even Numbers</title>
5   </head>
6   <body>
7     <h2>Even numbers in the array are:</h2>
8     <?php
9       $data = array(1, 3, 5, 2, 4, 5, 33, 31, 24, 54);
10      foreach ($data as $i) {
11        if ($i%2 == 0)
12          echo "$i ";
13      }
14      echo "<br>";
15    <?>
16  </body>
17 </html>
```

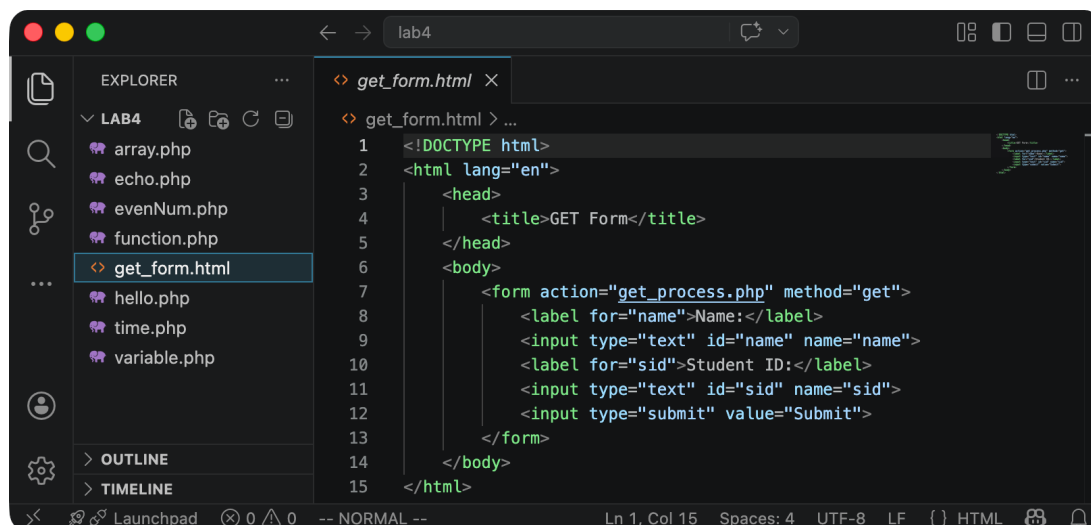
15. Access the webpage in your browser via URL: <http://localhost/can302/lab4/evenNum.php>



Even numbers in the array are:

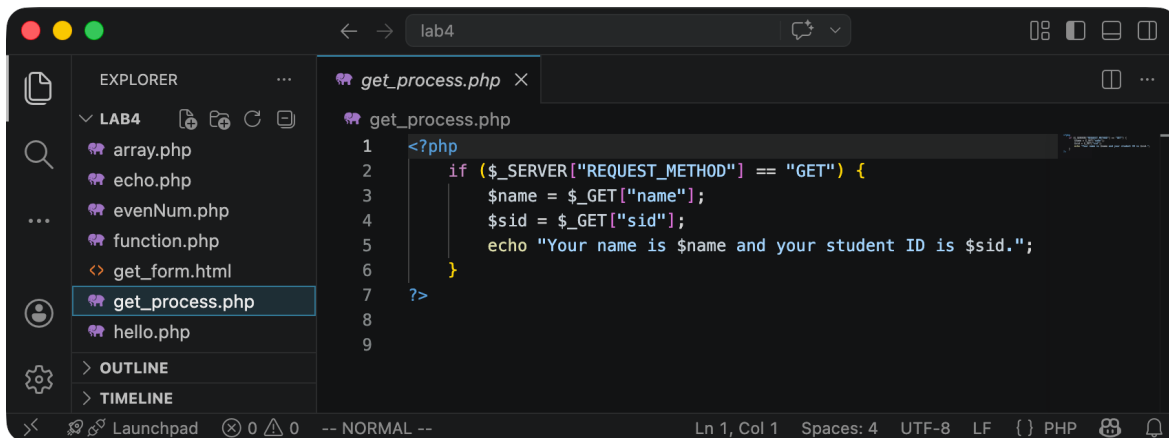
2 4 24 54

16. Create a new file "get_form.html" under "lab4" as follows:



```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>GET Form</title>
5   </head>
6   <body>
7     <form action="get_process.php" method="get">
8       <label for="name">Name:</label>
9       <input type="text" id="name" name="name">
10      <label for="sid">Student ID:</label>
11      <input type="text" id="sid" name="sid">
12      <input type="submit" value="Submit">
13    </form>
14  </body>
15 </html>
```

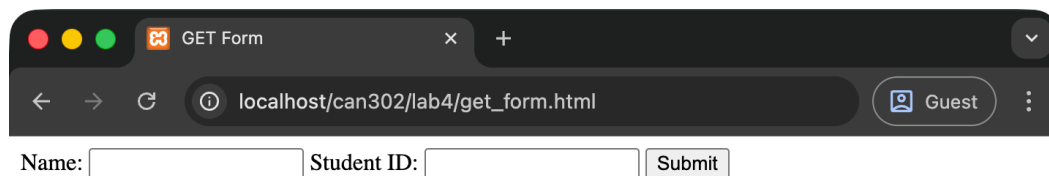
17. Create a new file "get_process.php" under "lab4" as follows:



The screenshot shows a code editor with a file explorer on the left. The file explorer shows a directory named 'LAB4' containing several files: array.php, echo.php, evenNum.php, function.php, get_form.html, get_process.php (selected), and hello.php. The main editor window shows the content of 'get_process.php' with the following PHP code:

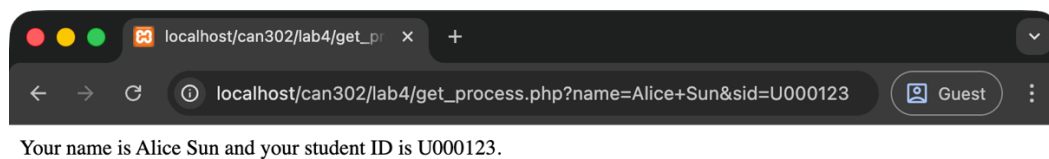
```
1 <?php
2     if ($_SERVER["REQUEST_METHOD"] == "GET") {
3         $name = $_GET["name"];
4         $sid = $_GET["sid"];
5         echo "Your name is $name and your student ID is $sid.";
6     }
7 ?>
```

18. Access the webpage in your browser via URL: http://localhost/can302/lab4/get_form.html



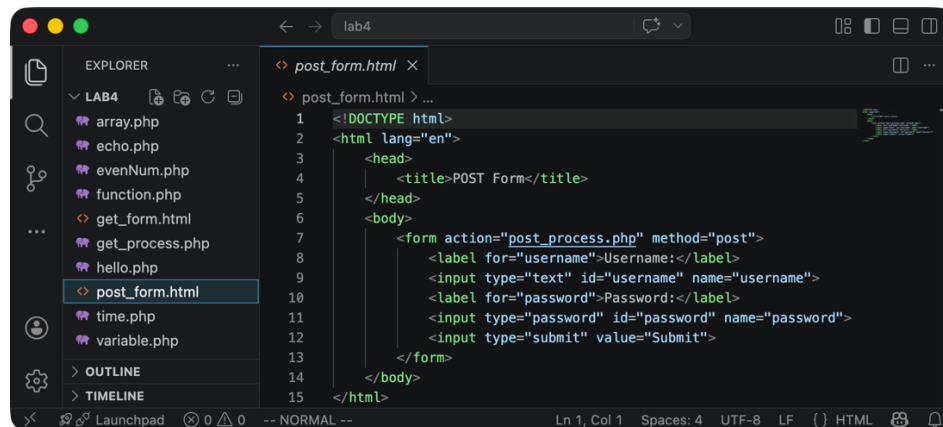
The screenshot shows a web browser window with the title 'GET Form'. The address bar shows the URL 'localhost/can302/lab4/get_form.html'. The page content displays a form with two input fields: 'Name:' and 'Student ID:', followed by a 'Submit' button.

19. Input a name and a student ID (i.e., Alice Sun and U000123), and click the "Submit" button:



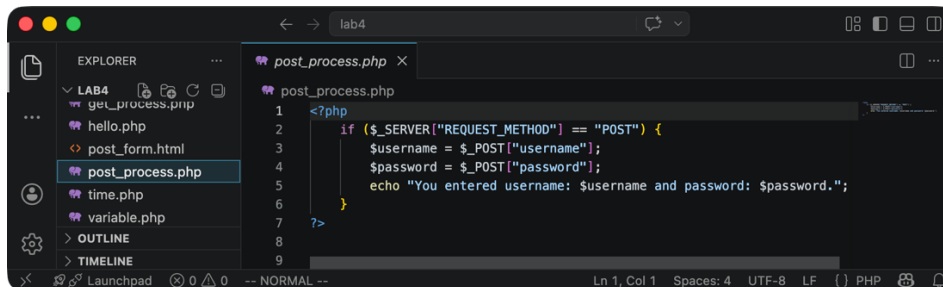
The screenshot shows a web browser window with the title 'localhost/can302/lab4/get_pr'. The address bar shows the URL 'localhost/can302/lab4/get_process.php?name=Alice+Sun&sid=U000123'. The page content displays the output of the PHP script: 'Your name is Alice Sun and your student ID is U000123.'

20. Create a new file "post_form.html" under "lab4" as follows:



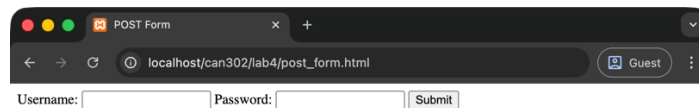
```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>POST Form</title>
</head>
<body>
  <form action="post_process.php" method="post">
    <label for="username">Username:</label>
    <input type="text" id="username" name="username">
    <label for="password">Password:</label>
    <input type="password" id="password" name="password">
    <input type="submit" value="Submit">
  </form>
</body>
</html>
```

21. Create a new file "post_process.php" under "lab4" as follows:

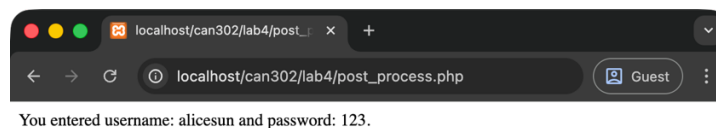


```
<?php
if ($_SERVER["REQUEST_METHOD"] == "POST") {
  $username = $_POST["username"];
  $password = $_POST["password"];
  echo "You entered username: $username and password: $password.";
}
```

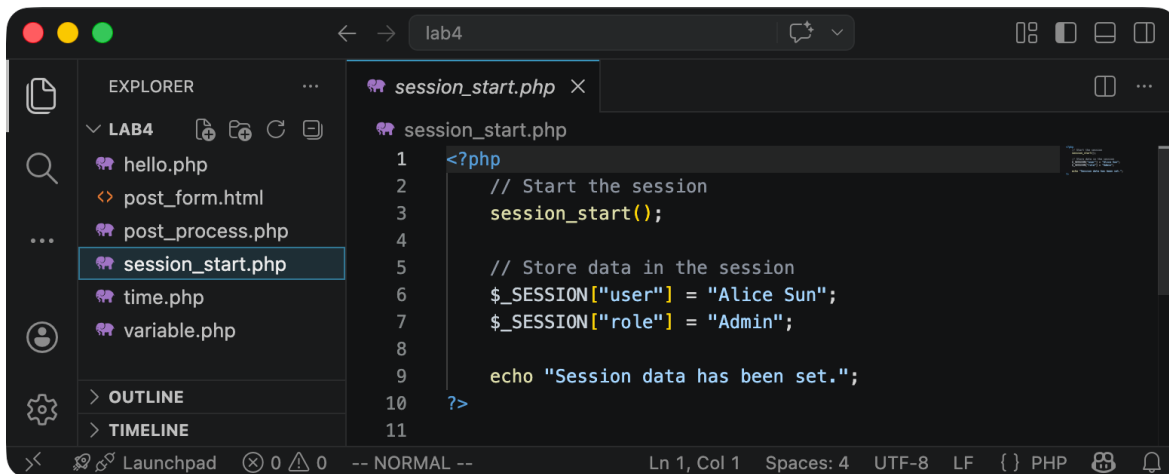
22. Access the webpage in your browser via URL: http://localhost/can302/lab4/post_form.html



23. Input a username and a password (i.e., alicesun and 123), and click the "Submit" button:



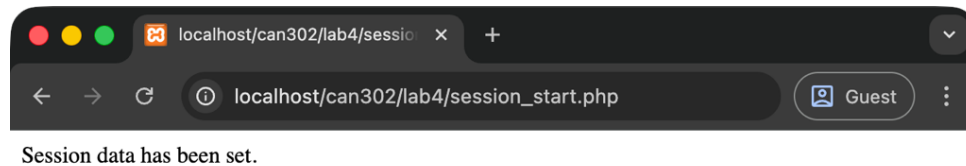
24. Create a new file "session_start.php" under "lab4" as follows:



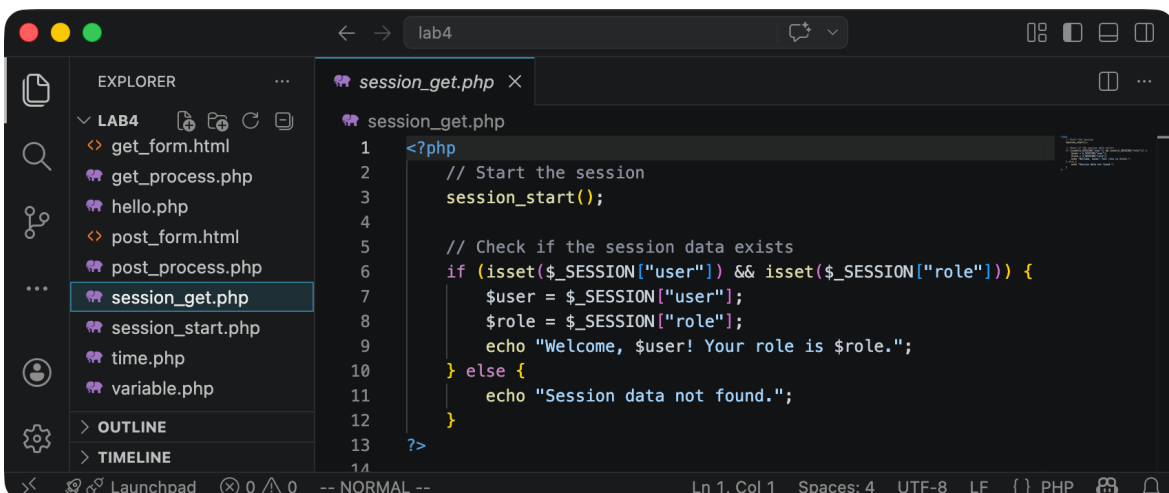
The screenshot shows a code editor with a file explorer on the left. The file explorer shows a directory named 'LAB4' containing several files: 'hello.php', 'post_form.html', 'post_process.php', 'session_start.php' (highlighted), 'time.php', and 'variable.php'. The main editor area shows the content of 'session_start.php' with the following code:

```
1 <?php
2 // Start the session
3 session_start();
4
5 // Store data in the session
6 $_SESSION["user"] = "Alice Sun";
7 $_SESSION["role"] = "Admin";
8
9 echo "Session data has been set.";
10 ?>
11
```

25. Access the webpage in your browser via URL: http://localhost/can302/lab4/session_start.php



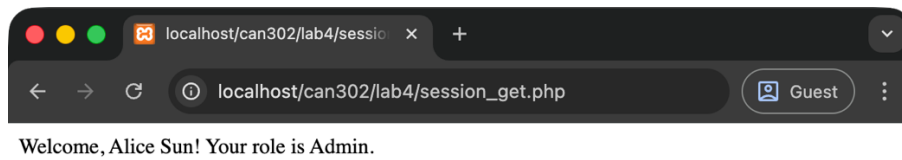
26. Create a new file "session_get.php" under "lab4" as follows:



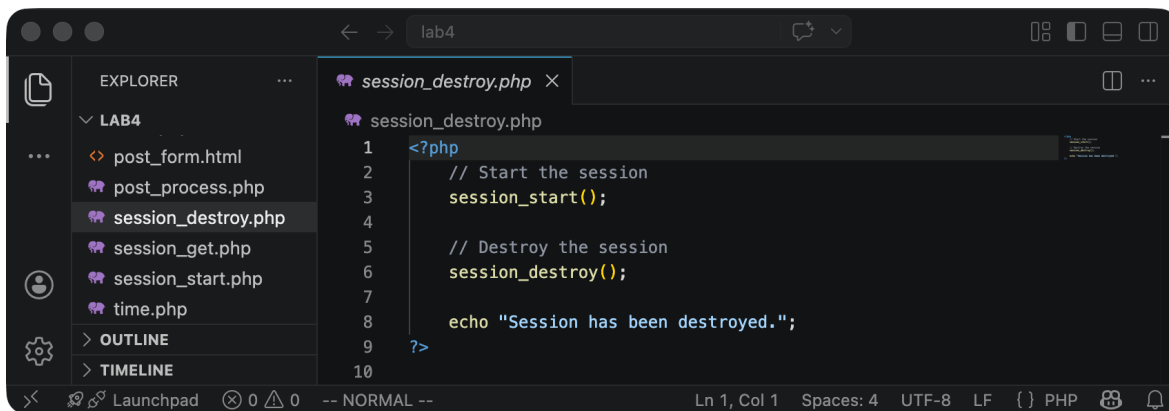
The screenshot shows a code editor with a file explorer on the left. The file explorer shows a directory named 'LAB4' containing several files: 'get_form.html', 'get_process.php', 'hello.php', 'post_form.html', 'post_process.php', 'session_get.php' (highlighted), 'session_start.php', 'time.php', and 'variable.php'. The main editor area shows the content of 'session_get.php' with the following code:

```
1 <?php
2 // Start the session
3 session_start();
4
5 // Check if the session data exists
6 if (isset($_SESSION["user"]) && isset($_SESSION["role"])) {
7     $user = $_SESSION["user"];
8     $role = $_SESSION["role"];
9     echo "Welcome, $user! Your role is $role.";
10 } else {
11     echo "Session data not found.";
12 }
13 ?>
14
```

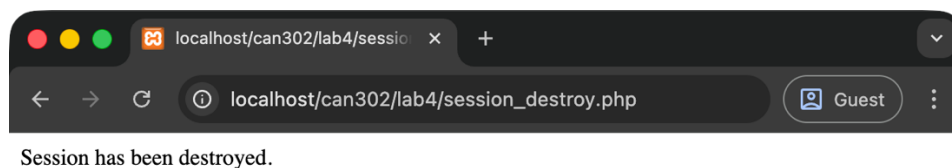
27. Access the webpage in your browser via URL: http://localhost/can302/lab4/session_get.php



28. Create a new file "session_destroy.php" under "lab4" as follows:

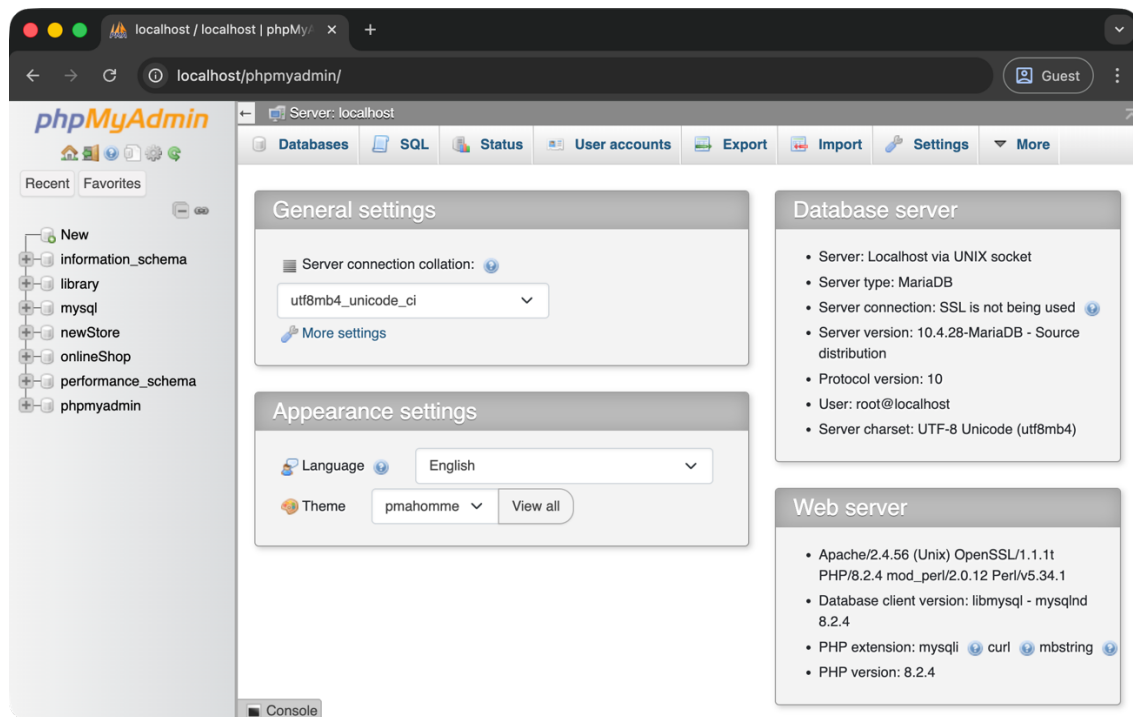


29. Access the webpage in your browser via URL: http://localhost/can302/lab4/session_destroy.php

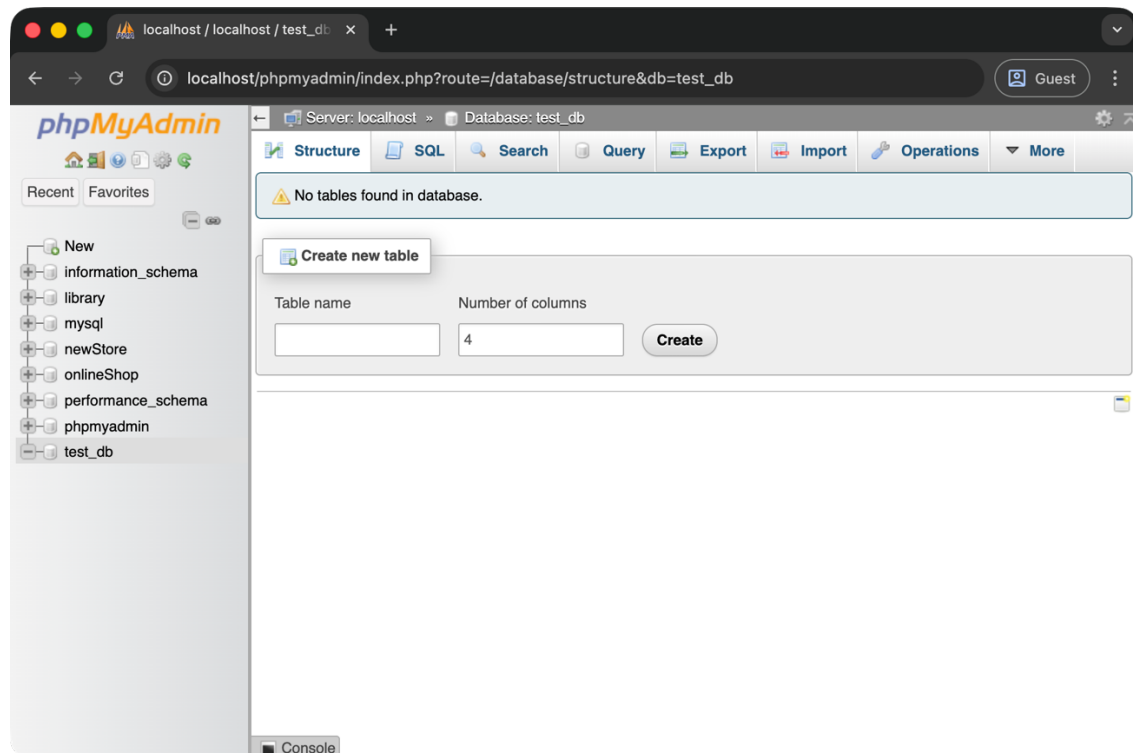


B. SQL Basics

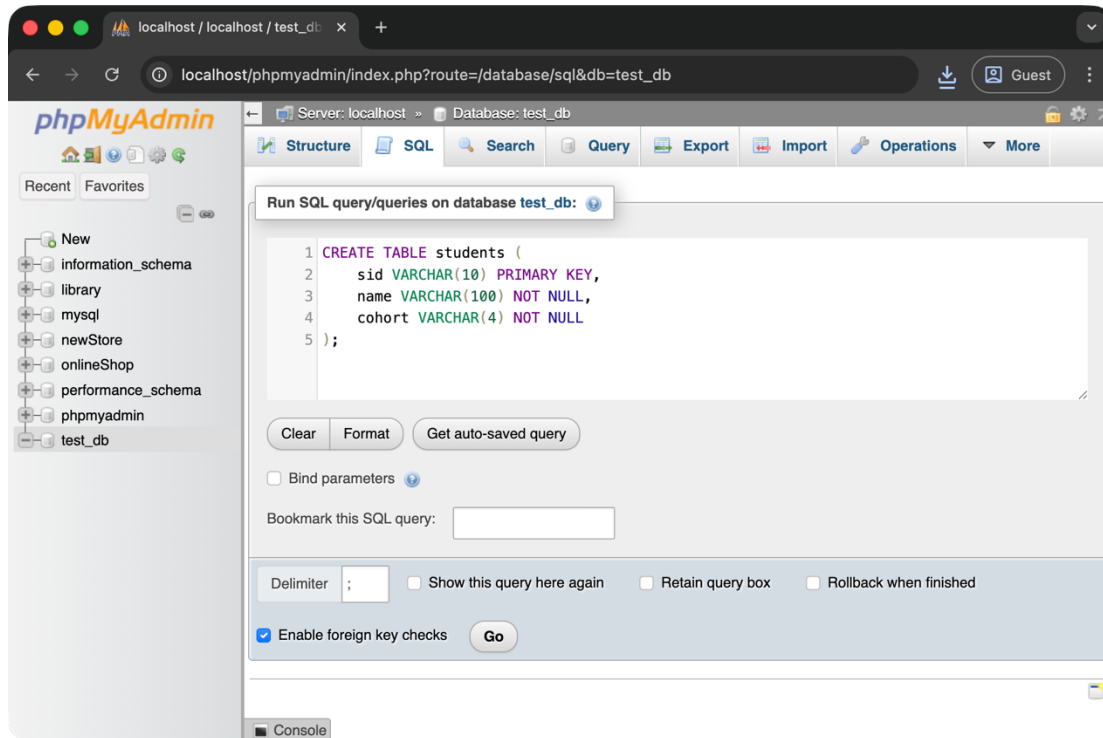
1. Open the browser and visit <http://localhost/phpmyadmin>



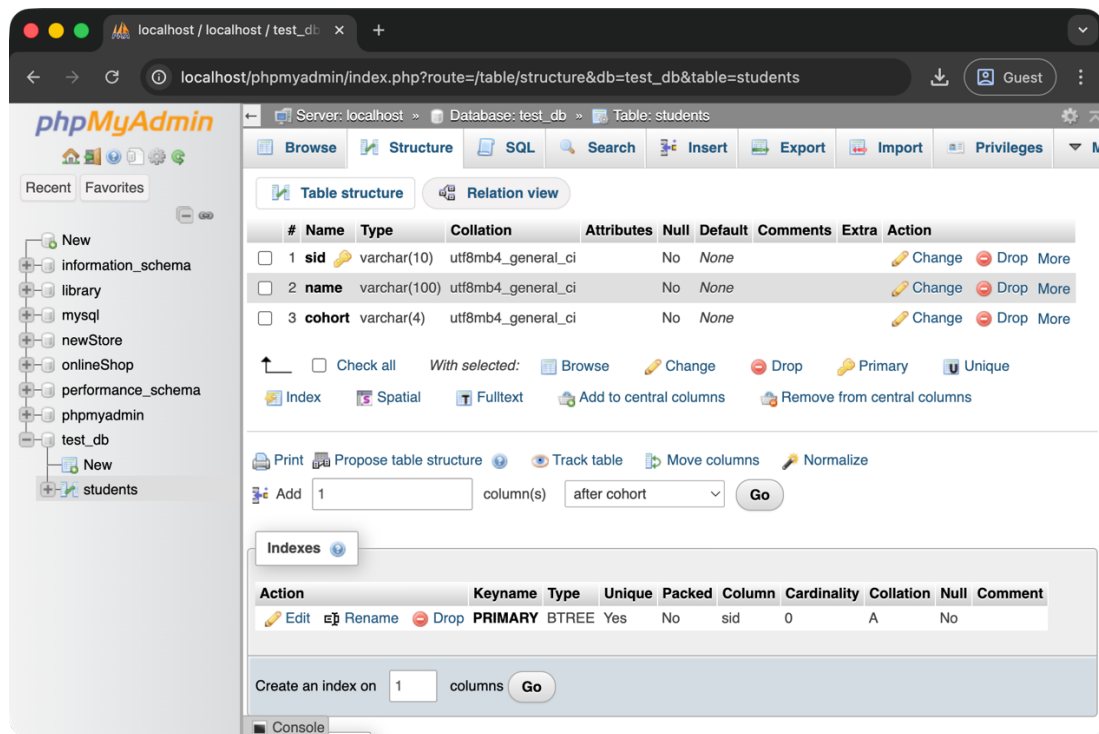
2. In the phpMyAdmin interface, click "New" in the left panel, enter "test_db" in the "Database" input box, and then click the "Create" button. Then, the "test_db" will appear in the left panel:



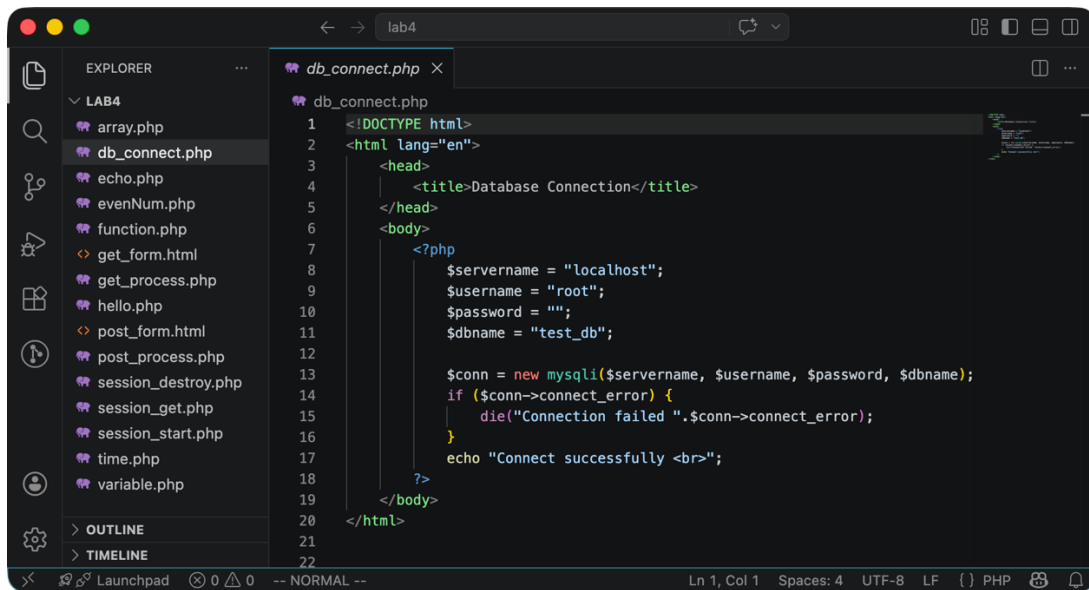
3. Select the newly created "test_db" database and click the "SQL" tab. Then, enter the following SQL statement in the text box, and click the "Go" button:



4. The statement creates a table named "students" with three fields: sid, name, and cohort:



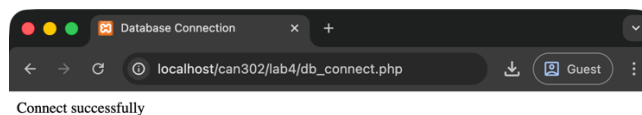
5. Create a new file "db_connect.php" under "lab4" as follows:



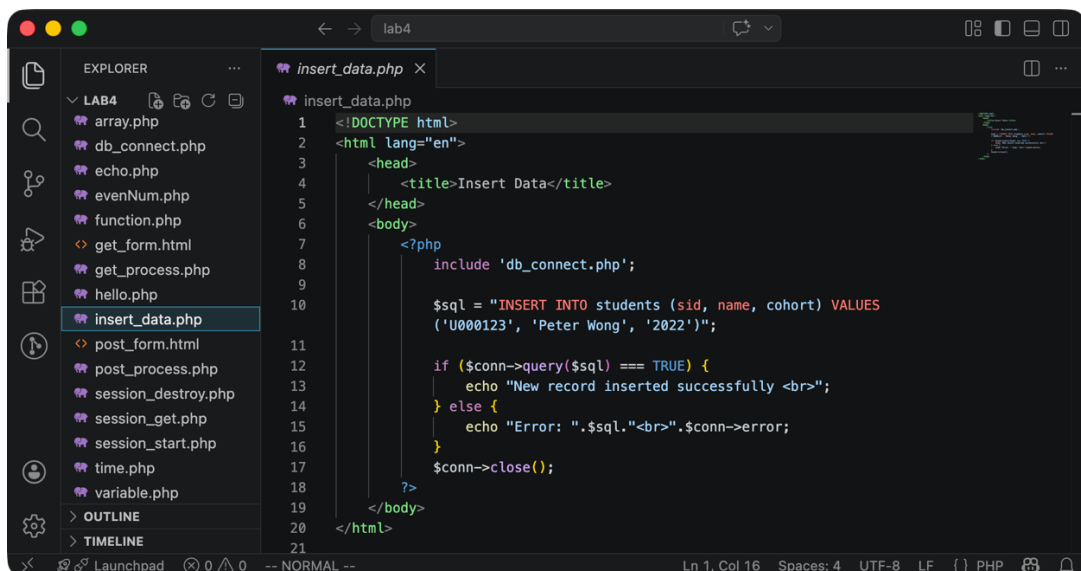
The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a directory named "LAB4" containing several files, including "db_connect.php". The code editor shows the contents of "db_connect.php", which is an HTML document with a title "Database Connection" and a PHP script that connects to a MySQL database. The PHP script sets the servername to "localhost", username to "root", password to "", and dbname to "test_db". It then uses the mysqli class to connect to the database. If the connection fails, it dies with an error message. If the connection is successful, it echoes "Connect successfully
".

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Database Connection</title>
5   </head>
6   <body>
7     <?php
8       $servername = "localhost";
9       $username = "root";
10      $password = "";
11      $dbname = "test_db";
12
13      $conn = new mysqli($servername, $username, $password, $dbname);
14      if ($conn->connect_error) {
15        die("Connection failed ".$conn->connect_error);
16      }
17      echo "Connect successfully <br>";
18    <?>
19  </body>
20 </html>
```

6. Open the browser and visit http://localhost/can302/lab4/db_connect.php



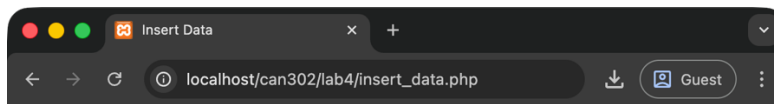
7. Create a new file "insert_data.php" under "lab4" as follows:



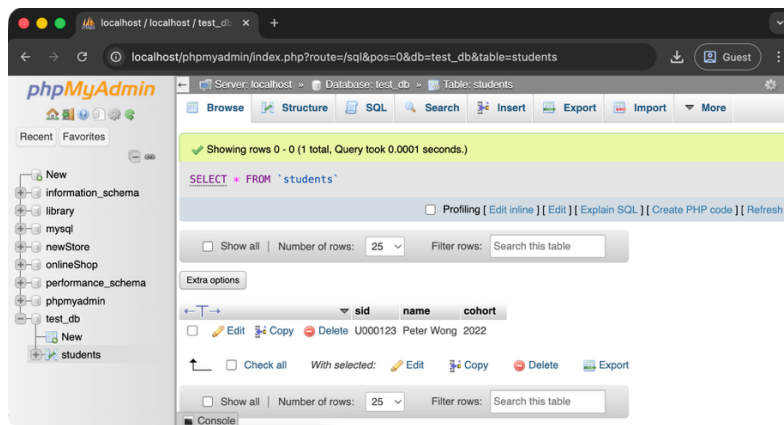
The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a directory named "LAB4" containing several files, including "insert_data.php". The code editor shows the contents of "insert_data.php", which is an HTML document with a title "Insert Data" and a PHP script that inserts data into a MySQL database. The PHP script includes "db_connect.php" and uses the mysqli class to connect to the database. It then uses the query method to execute an INSERT statement. If the query is successful, it echoes "New record inserted successfully
". If the query fails, it echoes an error message. Finally, it closes the database connection.

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Insert Data</title>
5   </head>
6   <body>
7     <?php
8       include 'db_connect.php';
9
10      $sql = "INSERT INTO students (sid, name, cohort) VALUES
11      ('U000123', 'Peter Wong', '2022')";
12
13      if ($conn->query($sql) === TRUE) {
14        echo "New record inserted successfully <br>";
15      } else {
16        echo "Error: ".$sql."<br>".$conn->error;
17      }
18      $conn->close();
19    <?>
20  </body>
21 </html>
```

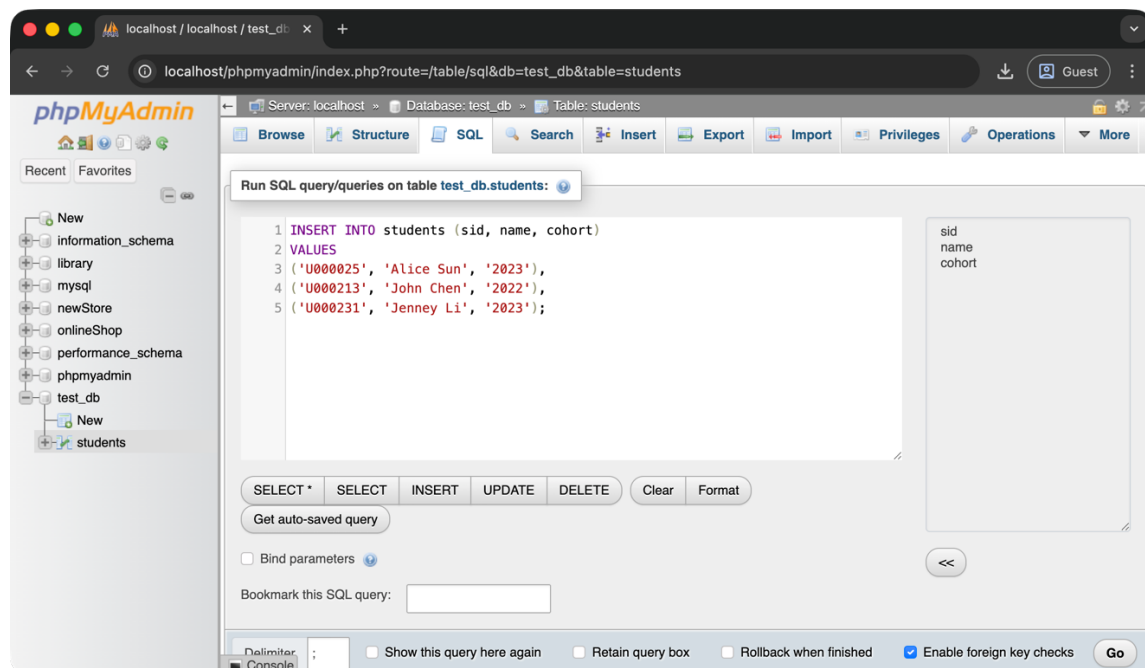
8. Open the browser and visit http://localhost/can302/lab4/insert_data.php



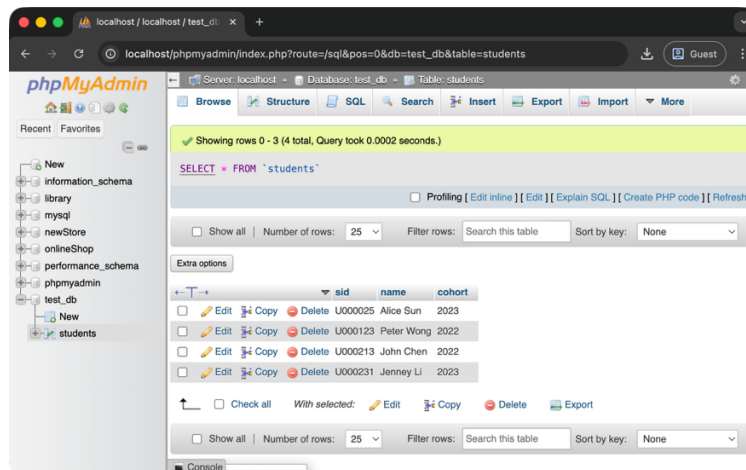
9. Visit <http://localhost/phpmyadmin>, and check the newly inserted record:



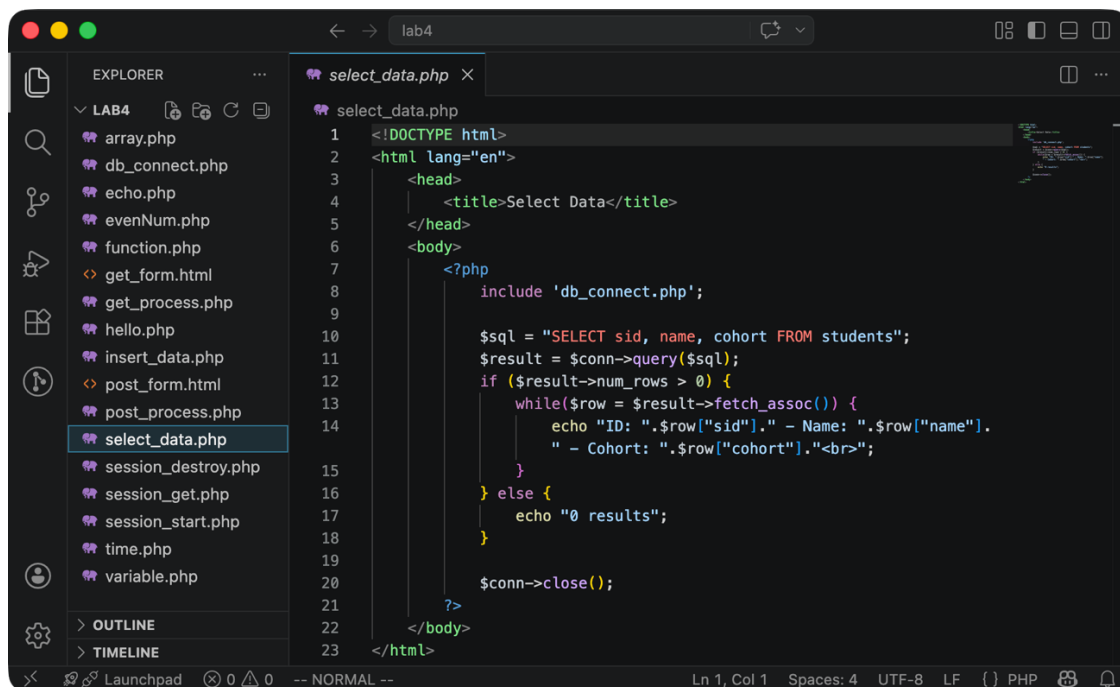
10. Insert more records in the table "students" as follows, and click the "Go" button:



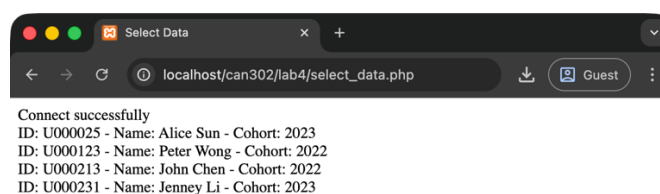
11. Check the newly inserted records:



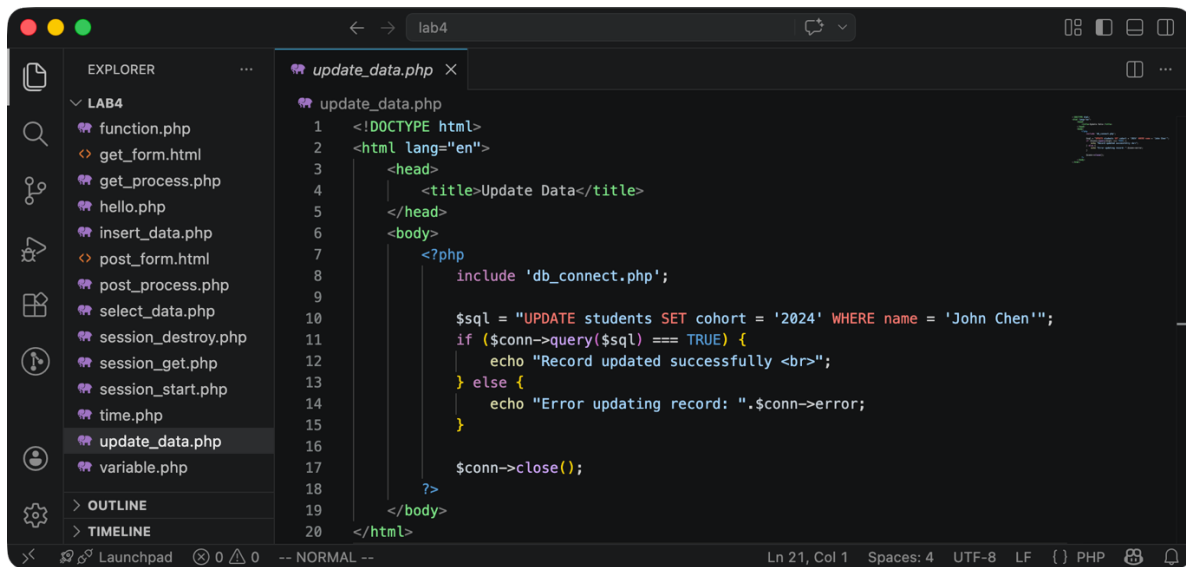
12. Create a new file "select_data.php" under "lab4" as follows:



13. Open the browser and visit http://localhost/can302/lab4/select_data.php

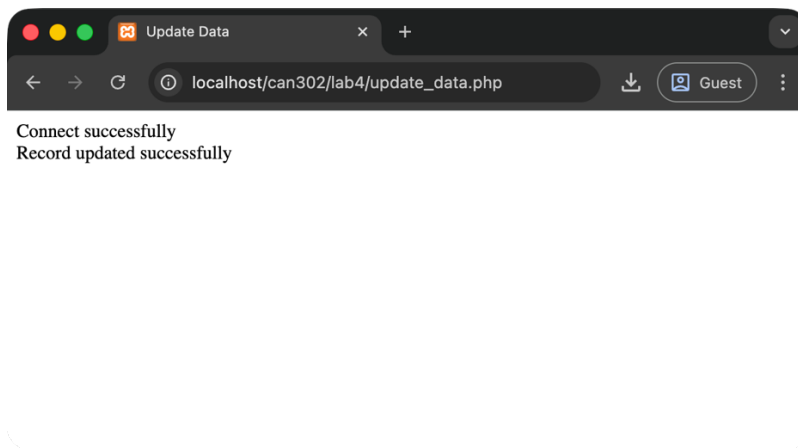


14. Create a new file "update_data.php" under "lab4" as follows:

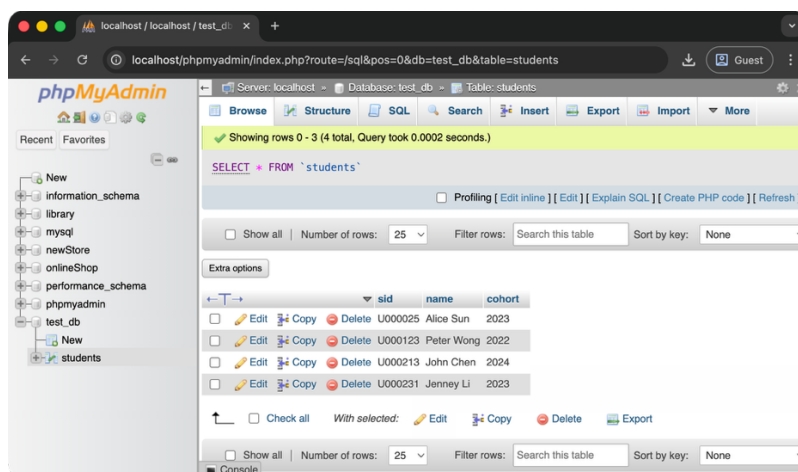


```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Update Data</title>
5   </head>
6   <body>
7     <?php
8       include 'db_connect.php';
9
10      $sql = "UPDATE students SET cohort = '2024' WHERE name = 'John Chen'";
11      if ($conn->query($sql) === TRUE) {
12        echo "Record updated successfully <br>";
13      } else {
14        echo "Error updating record: " . $conn->error;
15      }
16
17      $conn->close();
18    ?>
19   </body>
20 </html>
```

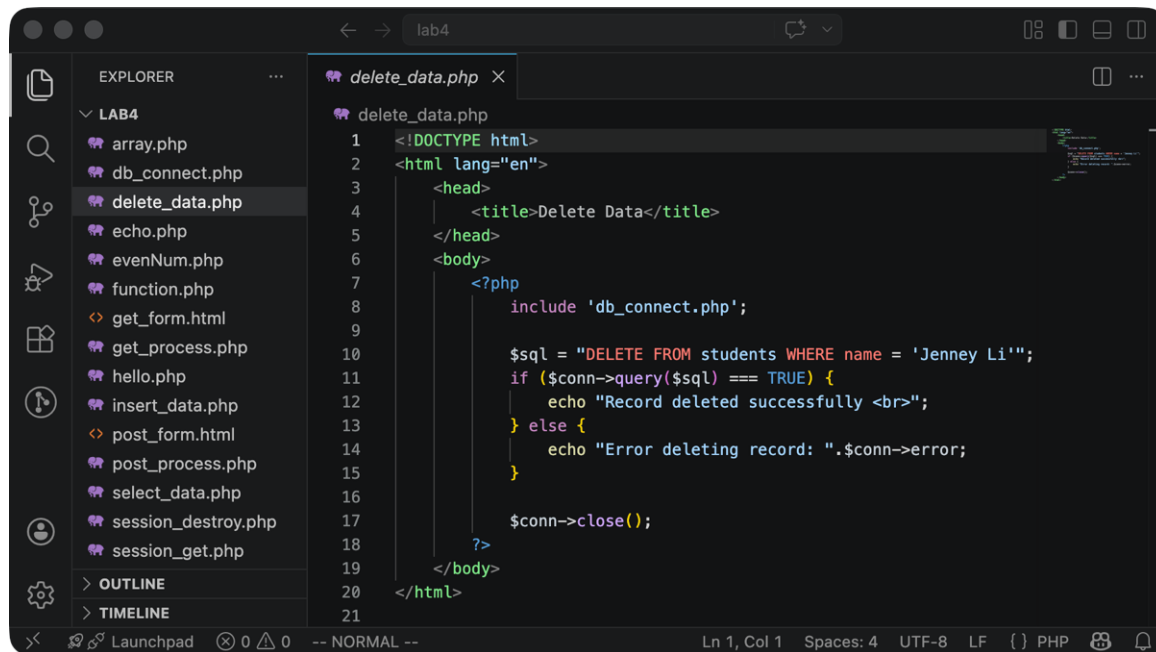
15. Open the browser and visit http://localhost/can302/lab4/update_data.php



16. Visit <http://localhost/phpmyadmin>, and check the updated record:

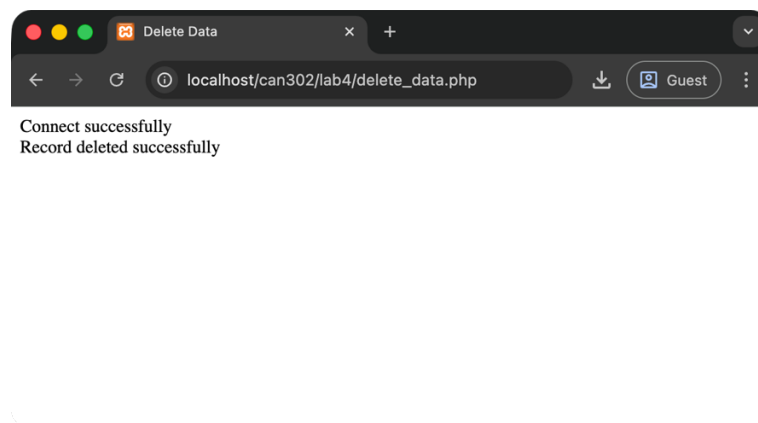


17. Create a new file "delete_data.php" under "lab4" as follows:

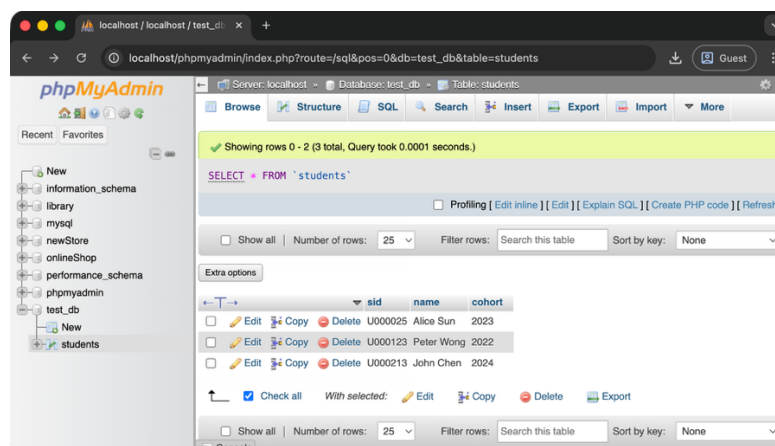


```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Delete Data</title>
5   </head>
6   <body>
7     <?php
8       include 'db_connect.php';
9
10      $sql = "DELETE FROM students WHERE name = 'Jenney Li'";
11      if ($conn->query($sql) === TRUE) {
12        echo "Record deleted successfully <br>";
13      } else {
14        echo "Error deleting record: ".$conn->error;
15      }
16
17      $conn->close();
18    ?>
19  </body>
20 </html>
21
```

18. Open the browser and visit http://localhost/can302/lab4/delete_data.php

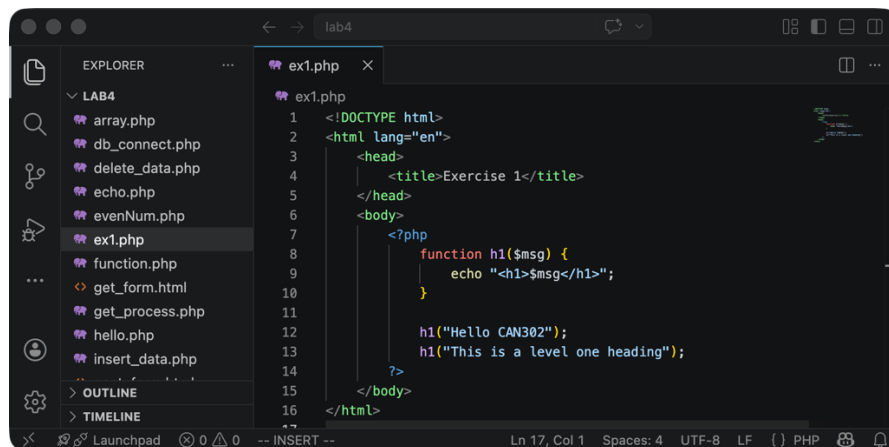


19. Visit <http://localhost/phpmyadmin>, and check if the record is deleted:

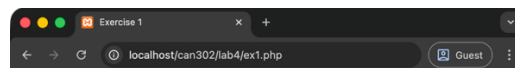


C. Hand-on Exercises

1. In this task, you are going to use function to simplify web page generation:
 - (a) Create a new file "ex1.php" under "lab4";
 - (b) Declare a function "h1" which accepts a string as argument and print out the string in HTML level 1 heading, i.e., <h1>...</h1>.



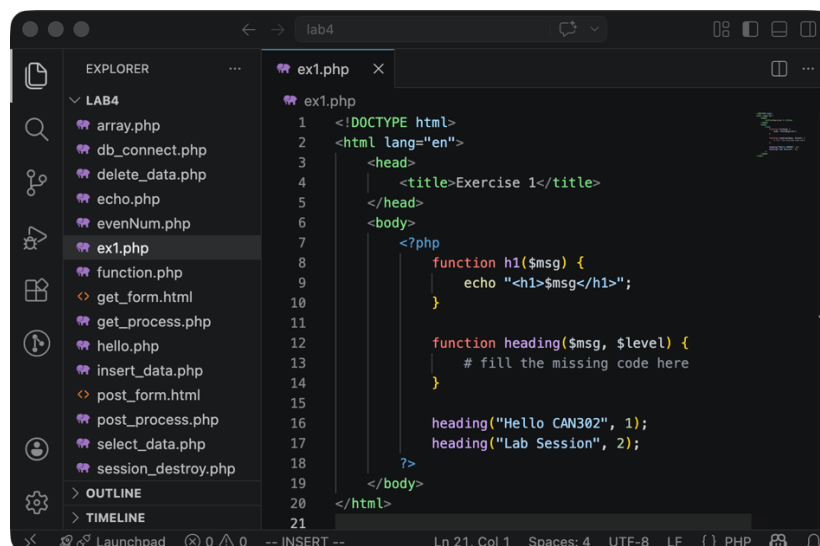
```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Exercise 1</title>
5   </head>
6   <body>
7     <?php
8       function h1($msg) {
9         echo "<h1>$msg</h1>";
10      }
11
12      h1("Hello CAN302");
13      h1("This is a level one heading");
14    ?>
15  </body>
16 </html>
```



Hello CAN302

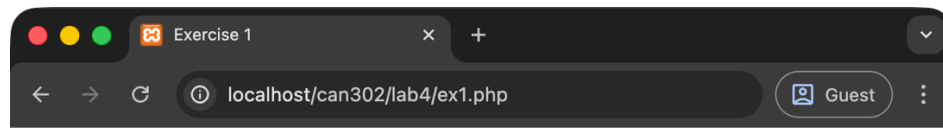
This is a level one heading

- (c) Declare another function call "heading" which accepts two arguments, a message and a heading level (you may assume the highest heading level is 6):



```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Exercise 1</title>
5   </head>
6   <body>
7     <?php
8       function h1($msg) {
9         echo "<h1>$msg</h1>";
10      }
11
12       function heading($msg, $level) {
13         # fill the missing code here
14      }
15
16       heading("Hello CAN302", 1);
17       heading("Lab Session", 2);
18    ?>
19  </body>
20 </html>
21
```

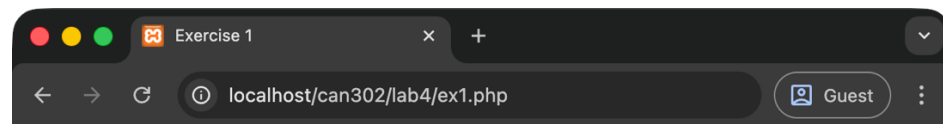
And it should look like:



Hello CAN302

Lab Session

(d) Try to generate the following page with for-loop and the function "heading":



Hello CAN302

Hello CAN302

Hello CAN302

Hello CAN302

Hello CAN302

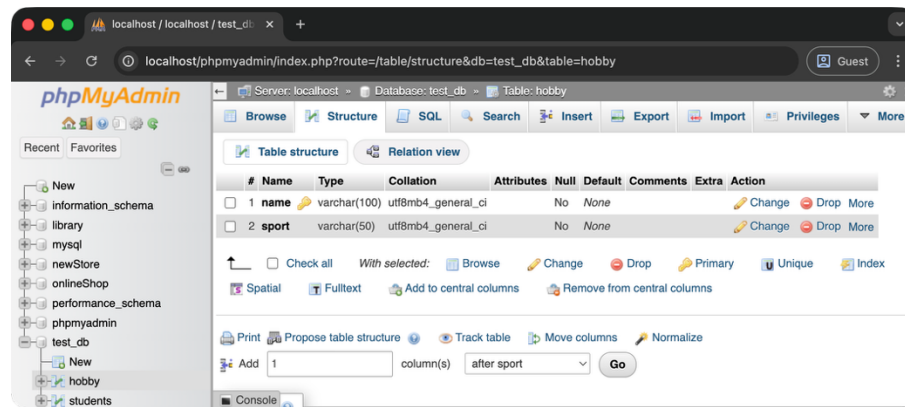
Hello CAN302

2. Practice database connection and table creation.

(a) Recall that we have created a database "test_db" and a table "students" with 3 columns:

sid	name	cohort
U000025	Alice Sun	2023
U000123	Peter Wong	2022
U000213	John Chen	2024

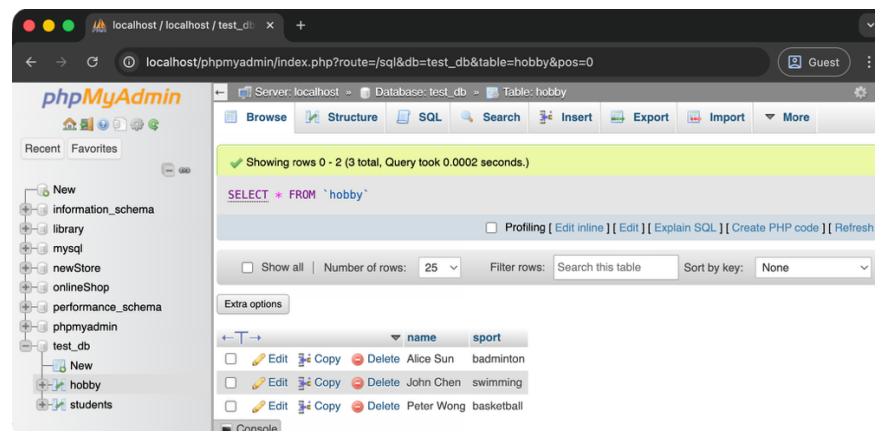
(b) Create a PHP script named "ex2_1.php" that connects to the "test_db" database and creates a table named "hobby" with columns "name" (VARCHAR (100) , primary key) and "sport" (VARCHAR (50)). After the creation, it should appear as follows:



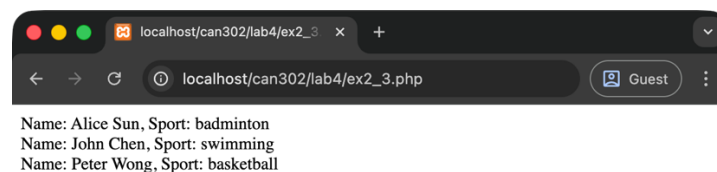
(c) Write a PHP script named "ex2_2.php" to insert three records as follows:

name	sport
Alice Sun	badminton
Peter Wong	basketball
John Chen	swimming

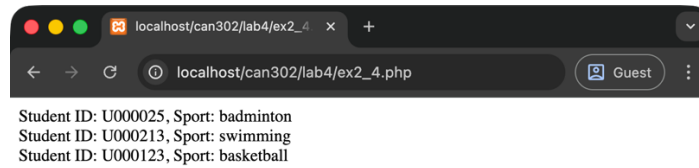
(d) After the insertion, it should appear as follows:



(e) Write a PHP script named "ex2_3.php" to query all records from the "hobby" table, and display their names and sports. It should appear as follows:

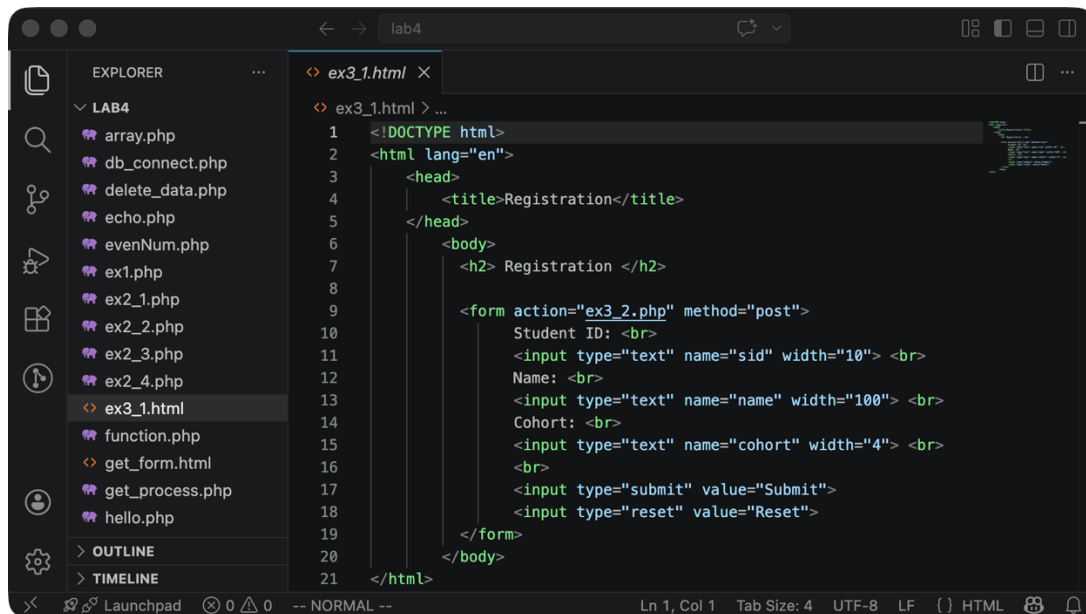


- (f) Write a PHP script named "ex2_4.php" to join the tables of "students" and "hobby", and display the student IDs along with their sports. It should appear as follows:

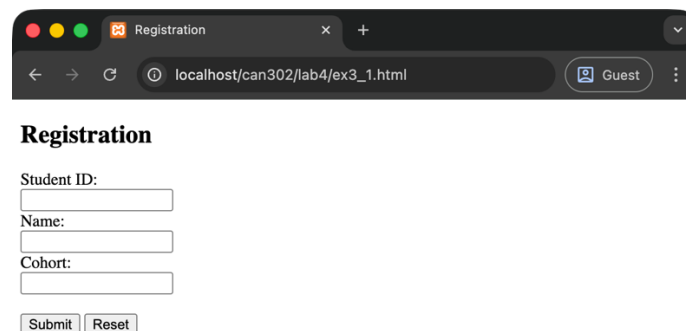


3. In this task, you are going to create a registration page with form and database.

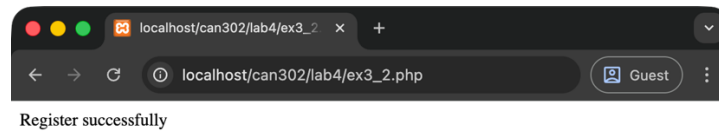
- (a) Create a new file named "ex3_1.html" as follows:



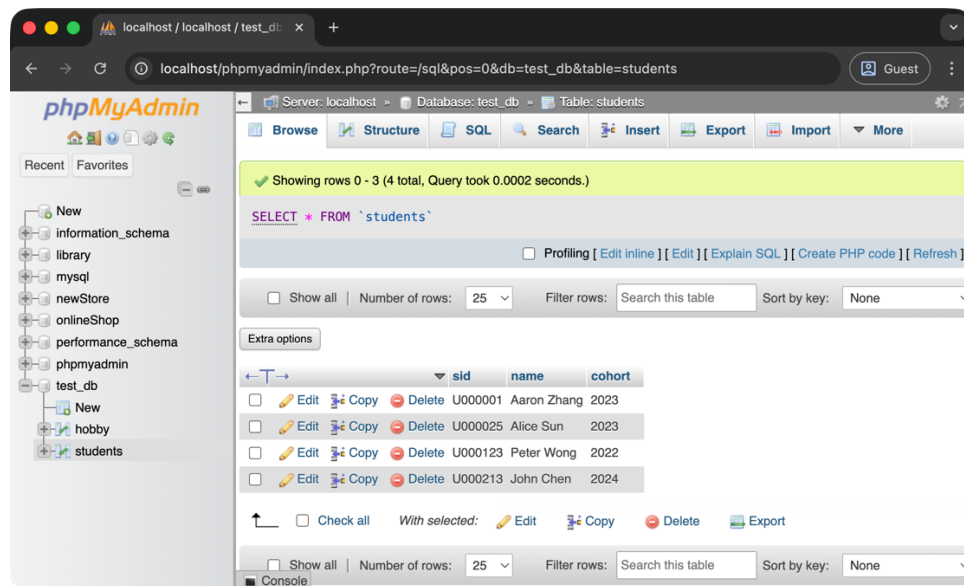
- (b) Open the browser and visit http://localhost/can302/lab4/ex3_1.html



- (c) Create a new file "ex3_2.php" to handle the form data and insert into the table "students" under "test_db".
- (d) Input a student ID, name, and cohort (e.g., U000001, Aaron Zhang, 2023), and click the "Submit" button:



- (e) Visit <http://localhost/phpmyadmin>, and check if the record is inserted:



Summary

We have achieved the following things in this lab session:

- Write server-side script programs in PHP
- Generate web pages content with functions
- Create tables in MySQL
- Access database in PHP